

PROJECT DESIGN PHASE

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S. No	Parameter	Description
1	Problem Statement (Problem to be solved)	Electricity consumption data is generated across multiple states and regions, but analyzing this large volume of data manually is difficult. Government departments and energy analysts often lack interactive tools to identify consumption trends, compare regional electricity usage, and support data-driven decision-making for efficient energy management.
2	Idea / Solution Description	The proposed solution uses an electricity consumption dataset stored in a MySQL database. SQL queries are used for data retrieval and analysis, while Tableau Desktop is used to build interactive dashboards and stories showing state-wise usage, monthly trends, region-wise comparison, top electricity consumers, and business insights. These dashboards are published on Tableau Public and embedded into a Flask web application, allowing users to analyze electricity consumption through an easy-to-use web interface.
3	Novelty / Uniqueness	Instead of traditional reports, the solution provides interactive dashboards with filters that allow users to explore electricity consumption data dynamically. Integrating MySQL, Tableau, Tableau Public, and Flask creates a complete analytics platform that improves visualization, accessibility, and decision-making for electricity management.
4	Social Impact / Customer Satisfaction	Electricity boards, government organizations, researchers, and analysts can better understand electricity demand patterns across different states and regions. This helps improve electricity distribution planning, optimize resource allocation, support sustainable energy management, and provide better services to consumers.
5	Business Model (Revenue Model)	The solution can be deployed as a subscription-based analytics platform for electricity boards, utility companies, and government organizations. Revenue can also be generated through customized dashboard development, enterprise analytics solutions, maintenance services, and consulting for energy management projects.

S. No	Parameter	Description
6	Scalability of the Solution	The solution can be expanded by integrating real-time electricity consumption data, smart meter information, IoT devices, AI-based electricity demand forecasting, renewable energy analytics, and support for district-wise and city-wise electricity monitoring without significant changes to the existing system architecture.